There is an important theorem that you can try to prove. I will call it

> **The Spin-Polarization Principle:** *Any state of a single spin is an eigenvector of some component of the spin.*

In other words, given any state

$$|A\rangle = \alpha_u|u\rangle + \alpha_d|d\rangle,$$

there exists some direction $\hat{n}$, such that

$$\vec{\sigma}\cdot\vec{n}\ |A\rangle = |A\rangle.$$

This means that for any spin state, there is some orientation of the apparatus $\mathcal{A}$ such that $\mathcal{A}$ will register +1 when it acts. In physics language, we say that the states of a spin are characterized by a *polarization vector*, and along that polarization vector the component of spin is predictably +1, assuming of course that you know the state-vector.

An interesting consequence of this theorem is that there is no state for which the *expectation values* of all three components of spin are zero. There is a quantitative way to express this. Consider the expectation value of the spin along the direction $\hat{n}$. Since |*A*> is an eigenvector of $\vec{\sigma}\cdot\vec{n}$ (with eigenvalue +1), it follows that the expectation value can be expressed as

$$\langle\vec{\sigma}\cdot\vec{n}\rangle = 1.$$

On the other hand, the expectation value of the perpendicular components of *σ* are zero in the state |*A*>. It follows that the squares of the expectation values of all three components of *σ* sum to 1. Moreover, this is true for any state:

$$\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1. \qquad (3.27)$$